

Warm-Up: Prerequisite Review — Answer Key

1. Evaluate each trigonometric expression.

(a) $\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$

(b) $\cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$

(c) $\tan\left(\frac{\pi}{4}\right) = 1$

(d) $\cos\left(\frac{5\pi}{6}\right) = -\frac{\sqrt{3}}{2}$

2. The table below shows some values of a function f .

x	0	$\frac{\pi}{6}$	$\frac{\pi}{2}$	$\frac{5\pi}{6}$	π
$f(x)$	0	$\frac{1}{2}$	1	$\frac{1}{2}$	0

(a) Is f a one-to-one function? **No**

(b) Explain.

The output $\frac{1}{2}$ corresponds to two different inputs ($\frac{\pi}{6}$ and $\frac{5\pi}{6}$), so the function fails the one-to-one condition.

3. Domain and range of $f(x) = \sin x$ (unrestricted).

Domain: **all real numbers** Range: $[-1, 1]$

4. For a function to have an inverse, it must be **one-to-one (injective)**. Graphical test: **the Horizontal Line Test — every horizontal line intersects the graph at most once.**